

VARUN KUMAR S

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EDUCATION

A J Institute of Engineering and Technology, Mangaluru

2022–2026

Bachelor of Engineering in Computer Science and Engineering

CGPA: 7.92 / 10.0 (up to Semester V)

TECHNICAL SKILLS

Programming: C, C++, Java, Python, JavaScript, SQL

Frontend: HTML, CSS, React, Vite

Backend: Node.js, Express.js, PHP

Databases: MySQL, Firebase Firestore

ML & AI: TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV

Tools: Git, GitHub, Firebase Hosting, Google Colab, VS Code

PROJECTS

Multi-Model Crop Health Monitoring and Disease Detection using Ensemble Models

Ongoing

- Developing an academic crop disease detection system through comparative evaluation of machine learning and deep learning models.
- Training and evaluating MobileNetV2, ResNet50, DenseNet121, Vision Transformer (ViT), CNN-based models, and SVM classifiers on agricultural image datasets.
- Performing image preprocessing, transfer learning, model validation, confusion matrix analysis, and performance comparison using accuracy, precision, recall, and F1-score metrics.
- Using PlantVillage and rice leaf disease datasets to study classification performance across multiple disease categories.

Swagriha – Family Productivity Platform

Personal Project

- Designed, developed, and deployed a Progressive Web Application using React, Vite, Firebase Hosting, and Firestore.
- Implemented shared grocery management, group communication, expense tracking, monthly reports, task management, and calendar-based planning.
- Built real-time synchronization and authentication workflows using Firebase services.
- Actively deployed and used in a real-world household environment while continuously improving functionality and usability.
- Exploring future expansion into a multi-tenant platform architecture as a panned product identity.

NoiseScope – Crowdsourced Noise Monitoring Platform

- Designed and implemented the complete MVP including frontend, backend, database integration, authentication workflow, and visualization modules.
- Built a platform for capturing ambient noise observations and visualizing trends through dashboards and location-based data collection.
- Proposed for scalable deployment in campuses, public spaces, and urban noise awareness initiatives.

Jewellery Rental Business Management Platform

Ongoing Client Project

- Developing a custom web solution for a jewellery rental business to streamline inventory visibility and customer interactions.
- Responsible for application architecture, frontend implementation, backend integration, and core business logic.
- Collaborating with a project partner managing client communication and operational coordination.

MedixFlow – Clinic Appointment Management System

- Independently developed a clinic appointment management platform using PHP, MySQL, HTML, and CSS as a self-driven exploration project.
- Implemented authentication, appointment scheduling, CRUD operations, administrative dashboards, and relational database concepts.

DisasterLink – Disaster Information Coordination Platform

- Built a platform for organizing disaster-related information, incident reporting, and resource coordination.
- Focused on information flow, accessibility, and structured access to critical updates during emergency situations.

ACHIEVEMENTS & INNOVATION

- Selected for the final interview of the **BOLPU Regional Startup Initiative** and presented the solution before industry experts.
- **1st Place** – Institutional Startup Competition, AJIET (Team Vayorca).
- Contributed to the development of an Arduino-based prototype addressing loss detection challenges in deep-sea fishing net operations.